When Does Linear Causal Abstraction Work? Mapping the Boundary on the Grassmannian

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Abstract

Distributed Alignment Search (DAS) identifies low-dimensional linear subspaces that mediate model behavior under interchange interventions, implicitly assuming that the relevant causal variable lives on the Grassmannian manifold $\mathrm{Gr}(k, d)$. When does this assumption hold, and what happens when it fails?

We construct an atlas of 14 modular arithmetic operations and report four findings. **(1)** Operations partition into three sharp classes—always, stochastic, and never Grassmannian—governed by grokking: Grassmannian variables appear if and only if the model generalizes. **(2)** Linear DAS returns IIA = 0.0 on fully grokked modular addition, while memorized models achieve IIA ≥ 0.86 without geometric structure, demonstrating that IIA alone cannot distinguish genuine causal variables from lookup tables. **(3)** A Structured pi-SAE—a label-conditional sparse autoencoder with end-to-end intervention training—recovers nonlinear causal variables where DAS fails, achieving IIA = 0.97 on GPT-2 hypernymy at k=1 (DAS: 0.07). Unconstrained nonlinear baselines achieve perfect IIA vacuously via degenerate encoder-decoders; we introduce *intervention faithfulness* metrics that expose this failure mode. **(4)** Mechanistic analysis reveals that nonlinearity is necessary for grokking but attention is not (DMF + ReLU groks 18× faster than the full transformer); that Fourier structure emerges *after* generalization, not before it; that weight-space SVD predicts causal geometry without optimization (IIA = 0.999 vs. DAS's 0.150); and that the causal subspace identity is gauge freedom—10 seeds produce orthogonal subspaces that all support equally valid interventions.

1 Introduction

Mechanistic interpretability seeks to identify causally active, human-understandable variables inside neural networks. Causal abstraction formalizes this goal: a model variable is "causal" if interventions on its internal representation behave like interventions on a high-level algorithmic variable (Geiger et al., 2021). Distributed Alignment Search (DAS) operationalizes causal abstraction by learning a rotation $Q \in \mathbb{R}^{d\times k}$ that identifies a k-dimensional subspace of the residual stream. Swapping the in-subspace component between two inputs and measuring whether the model produces the corresponding counterfactual output yields Interchange Intervention Accuracy (IIA) (Geiger et al., 2023).

DAS has been applied to syntax (Geiger et al., 2023), arithmetic (Wu et al., 2023), and board games (Li et al., 2023), but every application embeds a structural assumption: the causal variable is a *linear* subspace—a point on the Grassmannian manifold $\mathrm{Gr}(k, d)$. When the true causal variable is linear, DAS recovers it. When it is not, DAS may still return a high-IIA subspace that reflects memorization or distributional artifacts rather than genuine causal structure.

We do not ask whether DAS "works" in the narrow sense of achieving high IIA—it almost always does. We ask: **when does the linear assumption hold?** Is the subspace DAS recovers a genuine Grassmannian point with the geometric properties expected of a linear causal variable, or an artifact of searching in the wrong function class? And when it fails, **can a nonlinear method recover the causal variable?**

To answer these questions, we construct an atlas of 14 operations over $\mathbb{Z}_p$ spanning the boundary between linear and nonlinear causal structure. For each, we train a single-layer transformer in the standard grokking setup, fit DAS at multiple subspace dimensions, and evaluate the recovered subspace with three diagnostics beyond IIA: equivariance under the operation's symmetry group, circle geometry of the DAS projection, and intrinsic dimension from k-sweeps. We then apply a Structured pi-SAE (Narayanaswamy et al., 2017) to the never-Grassmannian operations, showing that nonlinear causal variables exist even where linear methods fail.

The atlas reveals a sharp, grokking-governed partition into three classes: seven operations *always* produce Grassmannian variables, two are *stochastic* (same operation, different seeds, opposite outcomes), and five *never* do. Grokked modular addition returns IIA = 0.0 under linear DAS despite having learned the correct algorithm, while memorized models achieve IIA $\geq$ 0.86 without geometric structure—demonstrating that IIA alone cannot distinguish genuine causal variables from lookup tables. A Structured pi-SAE recovers nonlinear causal variables where DAS fails, achieving IIA = 0.97 on GPT-2 hypernymy at k=1 (DAS: 0.07).

Beyond the atlas, we ask **how does grokking create causal structure?** Architecture ablations show that nonlinearity is necessary but attention is not. Fine-grained trajectory analysis reveals that Fourier structure emerges *after* generalization, and that weight-space SVD predicts causal geometry without optimization. The causal subspace itself is gauge freedom—the model does not prefer a canonical subspace.

Contributions.

1. **Atlas and stochastic grokking.** We construct an atlas of 14 operations mapping the boundary of linear causal abstraction. We discover stochastic grokking—seed-dependent outcomes for operations near the boundary—providing the cleanest evidence that generalization, not architecture, enables Grassmannian causal variables.
2. **IIA insufficiency and faithfulness metrics.** We show that high IIA is compatible with memorization (IIA = 1.0 at k=4 for non-grokking operations) and with degenerate encoder-decoders (NL-DAS achieves IIA = 1.0 via lookup tables). We introduce equivariance testing, intervention diversity ratio, and continuous distributional measures (KL, JS) as necessary complements.
3. **Structured pi-SAE.** A label-conditional sparse autoencoder with end-to-end intervention training recovers nonlinear causal variables where DAS fails entirely, with iVAE identifiability guarantees. Unconstrained nonlinear baselines are vacuous; the ELBO prevents degeneracy.
4. **Mechanistic analysis.** Nonlinearity is necessary for grokking but attention is not (DMF + ReLU groks 18$\times$ faster). Fourier structure lags generalization by $\sim$1,000 epochs. Weight-space SVD achieves IIA = 0.999 without optimization. The causal subspace identity is gauge freedom (10 seeds, pairwise overlap $\approx$ 0.008). Spectral profiling works on toy models but fails under superposition.
5. **Evaluation hierarchy.** We propose a four-level framework—interchange effectiveness, interchange faithfulness, distributional fidelity, and structural diagnostics—for evaluating causal variable claims.

2 Background

2.1 Causal abstraction and DAS

Let $h(x) \in \mathbb{R}^d$ be an intermediate activation for input x. DAS searches for an orthonormal $Q \in \mathbb{R}^{d\times k}$ whose column span defines a causal subspace. Given a base input x_b and source input x_s, the interchange intervention replaces the in-subspace component:

$$h' = h_b - QQ^\top h_b + QQ^\top h_s. \tag{1}$$

IIA measures the fraction of interventions for which the model's output matches the target counterfactual. High IIA indicates causal sufficiency: the subspace contains enough information to drive behavior under interchange. But sufficiency is not validity. A subspace may score high IIA because it encodes a memorized lookup table, because correlated directions redundantly encode the same label, or because the intervention distribution is too easy. Makelov et al. (2024) demonstrated this concretely: subspace activation patching can produce interpretability illusions where dormant pathways inflate IIA without reflecting genuine causal structure. Our central claim is that IIA must be paired with geometric diagnostics that test whether the subspace is a genuine linear causal variable rather than an artifact of the linear function class.

2.2 The Grassmannian manifold

The Grassmannian $\mathrm{Gr}(k, d)$ is the set of all k-dimensional linear subspaces of $\mathbb{R}^d$. A DAS solution $Q \in \mathbb{R}^{d\times k}$ represents a point $[Q] = \mathrm{span}(Q)$ on this manifold; any QR for $R \in O(k)$ represents the same point. The geodesic distance between two subspaces is determined by their *principal angles*: if Q_1, Q_2 have orthonormal columns, the singular values of $Q_1^\top Q_2$ are $\cos\theta_i$, and the geodesic distance is

$$d_{\mathrm{Gr}}([Q_1], [Q_2]) = \left(\sum_i \theta_i^2\right)^{1/2} \tag{2}$$

(Edelman et al., 1998; Absil et al., 2008). Every application of DAS is implicitly a search on this manifold. The question "does DAS work?" is really "does the true causal variable live on $\mathrm{Gr}(k, d)$?" When it does, DAS recovers a meaningful point. When it does not, DAS still returns a point on $\mathrm{Gr}(k, d)$, but that point may be an artifact.

2.3 Structured disentangled generative models

Semi-supervised deep generative models (Narayanaswamy et al., 2017) extend the VAE framework (Kingma & Welling, 2014) by partitioning the latent space into supervised and unsupervised factors. The generative model factors as $p(x, y, z) = p(x \mid y, z)\, p(y)\, p(z)$, where y is a labeled (interpretable) variable and z captures unlabeled nuisance variance. The recognition model $q(y, z \mid x) = q(z \mid x, y)\, q(y \mid x)$ uses neural network encoders, allowing nonlinear latent structure.

This framework is a natural generalization of DAS: DAS constrains the causal variable to live on $\mathrm{Gr}(k, d)$ (a linear subspace), while the structured VAE allows it to occupy a nonlinear manifold in activation space. The key advantage is disentanglement: the supervised factor y is trained to predict the operation's output, while the unsupervised factor z absorbs everything else. If the causal variable is linear, the VAE encoder should learn a rotation equivalent to DAS. If the causal variable is nonlinear (e.g., circular structure from Fourier representations), the VAE can recover it where DAS cannot.

The loss function combines the evidence lower bound (ELBO) with a supervised classification term:

$$\mathcal{L} = \underbrace{-\mathbb{E}_q[\log p(x \mid y, z)]}_{\text{reconstruction}} + \beta \underbrace{D_{\mathrm{KL}}(q(y, z \mid x) \,\|\, p(y)p(z))}_{\text{regularization}} + \alpha \underbrace{\mathcal{L}_{\mathrm{CE}}(y, \hat{y})}_{\text{supervision}}. \tag{3}$$

The encoder weights projecting to y provide a linearized approximation of the causal subspace; if this linearization has high overlap with the DAS subspace, it confirms that the causal variable is approximately Grassmannian.

2.4 Identifiability of the causal latent factor

A central concern for any latent-variable method is *identifiability*: does the encoder recover the *true* latent factors, or an arbitrary reparameterization? The Structured pi-SAE satisfies the conditions of iVAE (Khemakhem et al., 2020): (i) the task label y is an observed auxiliary variable; (ii) the label-conditional prior $p(z_{\mathrm{causal}} \mid y) = \mathcal{N}(\mu_y, \sigma_y^2 I)$ is an exponential family with per-label means that satisfy the rank condition; (iii) the ELBO reconstruction loss enforces an approximately injective decoder.

Proposition 1 (Causal variable recovery). *Under the iVAE conditions (Assumption 1 of Khemakhem et al. (2020)), the Structured pi-SAE encoder recovers the true causal variable up to a component-wise bijection: $\hat{z}_{causal} = \phi(z_{causal})$ where each ϕ_j depends only on z_j. The proof follows directly from Theorem 1 of Khemakhem et al. (2020); the supervised loss additionally resolves the permutation ambiguity.*

Operation	Formula	Category	Seeds tested
Multiplication	ab mod 113	Group action	42, 2024
Comp. addition	$(a+b)$ mod 91	Group action	orig, 42
Subtraction	$(a-b)$ mod 113	Group action	42
Division	a/b mod 113	Group action	42
Bitwise XOR	$a \oplus b$ mod 113	Group action	42, 137, 2024
Sum of squares	(a^2+b^2) mod 113	Polynomial	42
Cubic sum	(a^3+b^3) mod 113	Polynomial	42
Cubing	a^3 mod 113	Polynomial	42
Squaring	a^2 mod 113	Polynomial	42
Polynomial	(a^2+b) mod 113	Polynomial	42
Affine	$(2a+3b+5)$ mod 113	Polynomial	42
Max	$\max(a, b)$ mod 113	Piecewise	42
Abs. difference	$\|a-b\|$ mod 113	Piecewise	42
Power	a^b mod 113	Non-algebraic	42, 137

Table 1: Operation atlas. Group actions provide clean equivariance targets. Polynomials test whether nonlinear maps admit linear causal variables. Piecewise and non-algebraic operations probe the boundary.

Three consequences. (1) NL-DAS violates condition (iii): its decoder is not reconstruction-constrained, so multiple latent codes may map to the same activation and identifiability does not hold (§4.7). **(2) The component-wise ambiguity is harmless for IIA**: interchange swaps z_{causal} wholesale, so ϕ cancels. **(3) Linear DAS is the special case $\phi \in O(k)$.** When the true causal variable is linear, the structured encoder degenerates to a rotation, matching the DAS solution.

2.5 Grokking as a phase transition

Grokking is a training regime in which models memorize the training set long before generalizing (Power et al., 2022). Modular arithmetic models trained with weight decay exhibit grokking reliably for some operations but not others (Nanda et al., 2023; Zhong et al., 2023). Prior work analyzed grokking through Fourier representations and circular structure in the embedding space. Our perspective is complementary: we ask how the causal subspace recovered by DAS changes across the memorization-to-generalization transition, and whether that transition is deterministic or stochastic across random seeds. Liu et al. (2023) frame grokking as compression—moving from a memorized high-complexity solution to a generalizing low-complexity one; our Grassmannian perspective gives this compression a geometric interpretation as convergence to a specific point on $\mathrm{Gr}(k, d)$.

3 Methods

3.1 Operations and models

We study 14 binary operations over $\mathbb{Z}_p$ (Table 1), chosen to span group actions, polynomials, piecewise functions, and non-algebraic maps. For each operation, we train a single-layer transformer with $d_{\text{model}} = 128$, four attention heads ($d_{\text{head}} = 32$), and $d_{\text{MLP}} = 512$ on examples of the form $(a, b, =) \mapsto y$. Training uses the standard grokking setup: 30% train split, AdamW with weight decay 1.0, and 15,000–80,000 epochs depending on the operation. We classify a model as *grokked* if its final test cross-entropy is below 0.1.

3.2 DAS fitting and k-sweeps

For each trained model, we cache residual-stream activations and train a DAS rotation $Q \in \mathbb{R}^{d \times k}$ using interchange interventions for 400 gradient steps. We sweep $k \in \{2, 4, 6, 8, 10, 12, 16, 20, 24, 32\}$ to estimate the intrinsic dimension k^*, defined as the smallest k achieving IIA ≥ 0.95. Each k value is an independent

DAS optimization, not a nested subspace—k-sweep profiles reveal whether the causal variable is inherently low-dimensional or requires many dimensions.

3.3 Hard-example selection for IIA

Standard IIA evaluation includes many "easy" examples where the intervention does not change the model's top-1 prediction, inflating IIA regardless of whether the subspace is causally valid. We define a *hard example* as a (base, source) pair where: (1) the base and source have different correct outputs $y_b \neq y_s$, (2) the model's prediction matches the correct output for both base and source separately, and (3) the interchange intervention at the DAS subspace *can* flip the model's prediction from y_b to y_s. Hard IIA measures the fraction of hard examples where the flip actually occurs. This selection follows the methodology of CPCA-init DAS (Tower et al., forthcoming), which showed that standard IIA saturates too easily while hard IIA reveals genuine causal structure.

3.4 Equivariance testing

For operations with a natural group action, we test whether the DAS subspace respects the algebraic symmetry. Given input (a, b) with DAS projection $z = Q^\top h(a, b)$, we construct a shifted input $(a{+}1 \bmod p, b)$ and check whether the DAS projection rotates by $2\pi/p$ radians. The equivariant fraction is the proportion of test inputs satisfying this rotation property. Random orthogonal subspaces of the same dimension serve as controls; across all operations, random-subspace equivariance is below 1%.

3.5 Structured VAE for nonlinear causal variables

For each trained grokking model, we cache residual-stream activations at the post-MLP hook and train a structured VAE with:

- z_{causal}: k-dimensional latent factor ($k \in \{2, 4, 8\}$), semi-supervised with the operation's output label via a classification head.
- z_{nuisance}: m-dimensional unsupervised factor ($m = 15$) to absorb non-causal variance.
- Encoder: 2-layer MLP ($d_{\text{model}} \to 128 \to z$) with ReLU.
- Decoder: 2-layer MLP ($z \to 128 \to d_{\text{model}}$) with ReLU.

Training uses the combined ELBO + classification loss (Eq. 3) for 300 epochs with $\beta = 1.0$ and $\alpha = 10.0$.

VAE-based IIA. We evaluate the VAE's causal subspace by performing interchange interventions in the *latent* space: given base activation h_b and source activation h_s, we encode both, swap z_{causal} while keeping z_{nuisance} from the base, decode back to activation space, and measure whether the model produces the source's output. This is the nonlinear analogue of Eq. 1: instead of $QQ^\top$ projection, we use the encode-swap-decode path.

VAE equivariance. We test whether the VAE's z_{causal} exhibits equivariance under additive shifts, using the same protocol as for DAS (shift input $a \to a + 1$, check whether z_{causal} rotates consistently). Because the VAE encoder is nonlinear, it can represent circular structure directly in a 2D latent space.

3.6 Nonlinear DAS baseline

To isolate the contribution of the VAE's generative structure from the benefit of nonlinearity alone, we compare against an unconstrained nonlinear featurizer. Given MLP encoder $f : \mathbb{R}^d \to \mathbb{R}^d$ and decoder $g : \mathbb{R}^d \to \mathbb{R}^d$, the nonlinear DAS (NL-DAS) intervention is:

$$h' = g\big(f(h_b) - QQ^\top f(h_b) + QQ^\top f(h_s)\big) \tag{4}$$

where $Q \in \mathbb{R}^{d \times k}$ is learned jointly with f and g by maximizing interchange accuracy. Both f and g use the same architecture as the VAE encoder (2-layer MLP, 256 hidden units, ReLU), matching the VAE's representational capacity. Crucially, NL-DAS has *no reconstruction constraint*: f and g are trained exclusively on interchange loss and need not be (approximately) inverse to each other.

We also evaluate NL-DAS+recon, which adds a reconstruction penalty $\lambda \|g(f(h)) - h\|^2$ to the training loss, forcing f and g to approximate mutual inverses. This tests whether the unconstrained NL-DAS result degrades when the encoder-decoder cannot be degenerate.

3.7 Intervention faithfulness metrics

Binary IIA measures *interchange effectiveness*: does swapping the subspace component change the model's output to match the source? But effectiveness alone does not imply that the intervention is *faithful*—that it modifies only the targeted causal variable while preserving all other information. A method that achieves IIA = 1.0 by overwriting the entire activation with a class-specific template is effective but not faithful.

We introduce four complementary metrics to evaluate intervention faithfulness:

Continuous distributional metrics. Binary IIA saturates at 1.0 whenever the top-1 prediction flips, hiding differences in intervention quality. We measure the full distributional impact:

- **KL divergence**: $D_{\mathrm{KL}}(p_{\mathrm{clean}} \| p_{\mathrm{iv}})$, where p_{clean} and p_{iv} are the softmax distributions before and after intervention. Lower KL means the intervention preserves more of the clean distribution beyond the swapped variable.
- **Jensen-Shannon divergence**: the symmetric alternative $D_{\mathrm{JS}} = \frac{1}{2} D_{\mathrm{KL}}(p \| m) + \frac{1}{2} D_{\mathrm{KL}}(q \| m)$ where $m = (p+q)/2$. Bounded in $[0, \ln 2]$, more stable than KL when distributions have disjoint support.
- **Probability difference**: $p_{\mathrm{iv}}(y_s) - p_{\mathrm{iv}}(y_b)$, a continuous analog of binary IIA that captures the model's confidence margin, not just top-1 agreement.
- **Normalized logit difference**: $\frac{l_{\mathrm{iv}}(y_s) - l_{\mathrm{iv}}(y_b)}{|l_{\mathrm{clean}}(y_s) - l_{\mathrm{clean}}(y_b)|}$, where l denotes logits. A value of 1.0 means the intervention reproduces the clean model's margin exactly; values $\gg 1$ or $\ll 0$ indicate distortion.

Intervention diversity. For each source label y_s, we compute the standard deviation of the intervened activations h' across all base inputs with the same source. If the decoder produces the same h' regardless of which base input was used—varying only with y_s—the intervention is a lookup table: it does not preserve base-specific nuisance information. We define the *diversity ratio*:

$$\rho = \frac{\mathbb{E}_{y_s}\left[\mathrm{std}(h'_{y_s})\right]}{\mathbb{E}_{y_s}\left[\mathrm{std}(h_{y_s})\right]} \tag{5}$$

where the numerator is over intervened activations sharing source label y_s and the denominator is over the corresponding natural source activations. $\rho \approx 0$ indicates a lookup table (all intervened activations for the same y_s collapse to a single point). $\rho \approx 1$ indicates the intervention preserves the natural variation within each class.

Reconstruction fidelity. For encoder-decoder methods (NL-DAS, VAE), we measure $\|g(f(h)) - h\|^2$ on held-out activations. A high reconstruction MSE with high IIA is a warning sign: the decoder may be "hallucinating" the correct output rather than faithfully reconstructing the activation with a modified causal component. The generative constraint is not merely a regularizer—it is a necessary condition for identifiability (Proposition 1), as shown empirically by pi-VAE's failure despite having the correct prior direction (Table 3).

Operation	Grok	Loss	IIA(k=2)	k^*	Equiv.	Class
Always Grassmannian						
Multiplication	✓	0.000	1.000	2	0.995	Always
Subtraction	✓	0.000	1.000	2	1.000	Always
Division	✓	0.000	1.000	2	1.000	Always
Bitwise XOR	✓	0.003	1.000	2	1.000	Always
Cubic sum	✓	0.000	1.000	2	1.000	Always
Sum of squares	✓	0.005	1.000	2	0.987	Always
Max	✓	0.074	0.960	2	0.957	Always
Stochastic						
Comp. addition*	✓	0.000	—	—	0.998	Stochastic
Comp. addition (s42)	×	10.207	0.800	6	0.110	Stochastic
Power (s137)	✓	0.088	0.940	4	0.965	Stochastic
Power (s42)	×	1.143	0.980	2	0.665	Stochastic
Never Grassmannian						
Cubing	×	9.765	0.857	4	0.509	Never
Squaring	×	7.808	0.857	4	0.314	Never
Abs. difference	×	1.835	0.800	>32	0.189	Never
Polynomial	×	21.550	0.720	>32	0.015	Never
Affine	×	26.209	0.780	>32	0.026	Never

Table 2: Atlas results for 14 modular arithmetic operations. k^* is the smallest k achieving IIA $\geq$ 0.95. The "Always" class contains operations that grok and produce equivariance $>$ 95% across all tested seeds. *Original seed (unknown value); k-sweep not recorded.

3.8 Cross-distribution generalization

Standard train/test splits draw from the same distribution. For IOI on GPT-2, we additionally evaluate under distribution shift: (1) **disjoint names**—train on one set of proper names, evaluate on a completely disjoint set; and (2) **disjoint templates**—train on sentence templates 1–3, evaluate on templates 4–5 plus novel hard templates. A method that generalizes under distribution shift is finding a genuine causal variable; one that fails is exploiting distributional regularities in the training set.

4 Results

4.1 The atlas: a three-class partition

The 14 operations partition cleanly into three classes based on whether DAS recovers a Grassmannian causal variable (Table 2).

4.2 Linear DAS fails on grokked modular addition

The starkest evidence that the Grassmannian assumption can fail even for correct, generalizing models comes from modular addition. We train DAS at $k \in \{2, 4, 8, 16\}$ on residual-stream activations from a fully grokked model (test loss $< 10^{-7}$, test accuracy 100%). At *every* subspace dimension, DAS returns IIA = 0.0—not low, but exactly zero.

This is not a failure of DAS optimization. The model has learned the correct modular addition algorithm via Fourier representations (Nanda et al., 2023): the "identity" of each number a is encoded as $(\cos 2\pi fa/p, \sin 2\pi fa/p)$ for key frequencies f. This representation lives on S^1 (a circle embedded in $\mathbb{R}^d$), not on a linear subspace of $\mathbb{R}^d$. DAS searches $\mathrm{Gr}(k, d)$ for a flat plane that captures a curved manifold—a category error, not an optimization failure.

This motivates the Structured pi-SAE approach (§4.6): a nonlinear sparse encoder can map the circular representation to a structured latent space where interchange interventions succeed. On grokked addition, the Structured pi-SAE achieves IIA = 1.0 at k=2, confirming that the causal variable exists but is nonlinear.

4.3 Stochastic grokking: the same operation, opposite outcomes

The stochastic class provides the strongest evidence for our central claim. Power (a^b mod 113) grokked at seed 137 (test loss 0.088, equivariance 96.5%) but not at seed 42 (test loss 1.14, equivariance 66.5%). Composite addition ($a+b$ mod 91) grokked at one seed (test loss 0.000, equivariance 99.8%) but not at seed 42 (test loss 10.2, equivariance 11.0%). Same operation, same architecture, same hyperparameters—opposite outcomes from random initialization alone. Multi-seed stability experiments (10 seeds per stochastic operation) are underway and will be reported in the final version.

4.4 Memorization artifacts: high IIA, no structure

Squaring and cubing achieve IIA = 0.86 at $k = 2$ and IIA = 1.0 by $k = 4$ without grokking (test losses 7.8 and 9.8). Equivariance exposes the artifact: 31.4% for squaring and 50.9% for cubing, far below the >95% threshold. The lesson: **high IIA with low-k saturation is compatible with a lookup table**. "Works for interchange" and "is a Grassmannian causal variable" are different claims.

4.5 k-sweep profiles correlate with equivariance

Grassmannian variables saturate at IIA $\approx$ 1.0 by $k = 4$: their causal structure is inherently 2-dimensional. Non-Grassmannian variables show gradual IIA growth that may not reach 1.0 even at $k = 32$. Polynomial reaches IIA = 0.72 at $k = 2$ and requires $k \geq 8$ to approach 0.90.

4.6 Structured pi-SAE recovers nonlinear causal variables

We upgrade the vanilla structured VAE to a **Structured pi-SAE**: a label-conditional sparse autoencoder that partitions the latent space into z_{causal} (supervised, L_1-penalized) and z_{nuisance} (unsupervised). The "pi" denotes the label-conditional prior from Narayanaswamy et al. (2017); the "SAE" denotes L_1 sparsity on z_{causal} with an expansion factor of 8× (so k causal dimensions expand to $8k$ sparse features). This combines the identifiability guarantees of auxiliary supervision (Khemakhem et al., 2020) with the interpretability benefits of sparse dictionary learning (Bricken et al., 2023).

The 2×2 ablation. Both components—structured prior and sparsity—are necessary (Table 3). On grokking tasks: (1) the plain SAE (no label-conditional prior) achieves IIA = 1.0 on addition but degrades on harder operations (multiplication 0.72, quartic sum 0.32); (2) the pi-VAE (structured prior, no sparsity) achieves IIA $\approx$ 0 across all operations; (3) *only* the Structured pi-SAE achieves IIA = 1.0 uniformly across all grokked operations and IIA $\approx$ 0 on all non-grokked operations.

Reconstruction MSE as a falsifiability criterion. For non-grokked operations, reconstruction MSE is 11–24; for grokked operations, 0.6–1.2. High MSE with near-zero IIA means the model has *no structured causal variable to recover*—not that the method failed. This diagnostic is unavailable for NL-DAS, which returns IIA = 0.6–0.8 on non-grokked operations (false positives) because it has no reconstruction constraint.

End-to-end intervention training. For language model tasks where the causal structure is more complex, we add an **end-to-end (E2E) intervention CE loss**: after swapping z_{causal} and decoding, we run the intervened activation through the remaining layers and compute $-\log p(y_s \mid h')$. This differentiable objective directly optimizes for intervention success, analogous to how DAS trains on interchange accuracy. Training proceeds in two phases: 200 epochs of reconstruction + classification warmup, followed by E2E intervention training with an additive intervention $h' = h_b + \text{dec}(z_{\text{swap}}) - \text{dec}(z_{\text{orig}})$ that cancels reconstruction error.

Operation	Plain prior $\mathcal{N}(0, I)$ VAE	SAE	Structured prior $p(z \mid y)$ pi-VAE	**Str. pi-SAE**
Grokked				
Addition	0.02	1.00	0.00	**1.00**
Multiplication	0.00	0.72	0.00	**1.00**
Quartic sum	0.02	0.32	0.00	**1.00**
IOI (GPT-2)	0.56	0.83	0.95	**0.98**
Non-grokked (correct answer: ≈ 0)				
Mixed product	0.03	0.01	0.01	0.04
Squaring	0.00	0.00	0.00	0.00

Table 3: 2×2 ablation: IIA at k=2. Plain SAE works on addition but degrades on harder operations (multiplication 0.72, quartic sum 0.32) because without the structured prior, the sparse encoder has no target direction. pi-VAE (structured prior, no sparsity) achieves 0.00 on grokked operations—the prior provides the right direction but without sparsity the encoder spreads the causal signal across all dimensions, which cannot be cleanly swapped. Only Structured pi-SAE achieves 1.00 uniformly on all grokked operations and ≤ 0.04 on all non-grokked operations.

Method	**Hypernymy** k=1	k=4	**Gender bias** k=1	k=4
DAS	0.07	0.23	0.62	0.88
NL-DAS	0.72	0.92	1.00	1.00
Str. pi-SAE	0.58	0.52	0.41	0.52
Str. pi-SAE E2E	**0.97**	**0.98**	*pending*	*pending*

Table 4: Standard IIA on GPT-2 language tasks (layer 8, additive intervention). Structured pi-SAE with E2E training achieves 0.97 standard IIA on hypernymy at k=1, where DAS gets 0.07 and even NL-DAS gets only 0.72. Strict IIA (argmax = source token) is 0.90 at k=1.

Results on GPT-2 language tasks. We evaluate on five tasks from the MIB benchmark (Prakash et al., 2024) at layer 8 of GPT-2-small: gender bias, hypernymy, greater-than, SVA, and capitals. Table 4 reports the headline result.

On hypernymy, the Structured pi-SAE with E2E training achieves standard IIA = 0.97 at k=1—14× better than DAS (0.07) and 1.35× better than NL-DAS (0.72). Under the stricter argmax criterion (source token must beat all 50,000 vocabulary tokens), IIA remains 0.90. The E2E objective closes the gap between training and evaluation by directly optimizing the quantity we measure.

On the Indirect Object Identification task (Wang et al., 2023), DAS achieves IIA = 0.30 at k=2, consistent with the IOI causal variable spanning multiple attention heads. Structured pi-SAE achieves 0.98. NL-DAS achieves 1.00, but the diversity ratio $\rho \approx 0$ (vs. $\rho \approx 0.91$ for the pi-SAE) exposes it as vacuous (Table 5). The identifiability guarantee (Proposition 1) applies: each IOI sentence has a distinct indirect object label, the per-label means μ_y are well-separated, and the ELBO reconstruction constraint forces approximate decoder injectivity.

4.7 Unconstrained nonlinear DAS is vacuous

A natural response to DAS's linearity constraint is to prepend an MLP featurizer: learn a nonlinear coordinate system in which the causal variable *becomes* linear, then apply standard DAS. This approach—which we call NL-DAS (§3.6)—achieves IIA = 1.000 on IOI at $k = 1$, compared to DAS's 0.193. This looks like a dramatic improvement. It is not.

The lookup-table failure mode. NL-DAS is trained end-to-end on interchange loss with no reconstruction constraint. The encoder f and decoder g need not be approximate inverses, and the training objective

Task	k	DAS	NL-DAS	NL-DAS+r	VAE	ρ_{NL}	ρ_{VAE}
IOI	1	0.193	1.000	—	0.990	—	—
IOI	2	0.297	1.000	—	0.980	—	—
IOI	4	0.556	1.000	—	0.978	—	—
Addition	1	0.006	0.194	—	0.000	—	—
Addition	4	0.289	0.883	—	0.089	—	—
Mult.	4	0.344	0.961	—	0.000	—	—
Squaring	$*$	0.000	0.011	—	0.000	—	—

Table 5: NL-DAS achieves high IIA by learning degenerate encoder-decoders. The VAE column reports Structured pi-SAE results. ρ is the intervention diversity ratio (Eq. 5); NL-DAS+r adds a reconstruction penalty. — marks values pending from hard-mode experiments. All controls (random labels, reconstruction-only, untrained, unconstrained plain VAE) achieve IIA < 0.09, confirming the Structured pi-SAE is not cheating.

provides no incentive for them to be. This creates a degenerate solution: f maps every activation to a space where the first coordinate encodes the class label and the remaining coordinates are arbitrary; g ignores the base-specific content entirely and produces a "canonical activation for class y_s" regardless of which base input was used.

The intervention diversity ratio (§3.7) exposes this failure. Table 5 reports results for IOI and three grokking operations across $k \in \{1, 2, 4\}$.

Three lines of evidence confirm NL-DAS is vacuous:

1. **Controls rule out the structured VAE cheating.** Random-label VAE (IIA ≤ 0.02), reconstruction-only VAE with $\alpha = 0$ (IIA ≤ 0.005), untrained VAE (IIA $= 0.000$), and an unconstrained plain VAE with no causal/nuisance split (IIA ≤ 0.08) all fail. The structured VAE's success requires correct supervision *and* the causal-nuisance partition. This is the empirical counterpart of Proposition 1: NL-DAS fails Assumption 1(iii) because its decoder is not injective, so the iVAE identifiability guarantee does not apply (Khemakhem et al., 2020). Without inductive biases, unsupervised disentanglement is impossible (Locatello et al., 2019); the structured prior and reconstruction constraint are not optional regularizers but necessary conditions.

2. **NL-DAS+recon degrades.** When forced to reconstruct its input ($g(f(h)) \approx h$), NL-DAS can no longer achieve degenerate solutions. Results with the reconstruction penalty are forthcoming.

3. **Cross-distribution generalization.** NL-DAS performance under distribution shift (disjoint names and novel templates), where the encoder cannot memorize specific activation patterns from training, will be reported in the final version.

Why the VAE is not vacuous. The structured VAE's ELBO provides three constraints that prevent the lookup-table failure mode: (1) the reconstruction loss forces the decoder to produce activations close to the original, not class-specific templates; (2) the KL regularization prevents the encoder from collapsing the latent space to a few discrete points; and (3) the causal-nuisance partition ensures that z_{nuisance} absorbs base-specific information, so the decoder must use it. Together, these constraints mean that swapping z_{causal} while preserving z_{nuisance} genuinely modifies only the causal component.

Implications. Any nonlinear method for causal abstraction must be evaluated with intervention faithfulness metrics, not just IIA. The NL-DAS failure mode is not specific to our architecture—it applies to any unconstrained encoder-decoder trained exclusively on interchange loss. Prior work proposing neural network featurizers for causal abstraction (Geiger et al., 2024) should be re-evaluated with diversity ratio and reconstruction checks.

Architecture	Grok epoch	Fourier align.	Groks?
DMF + ReLU	∼2,000	0.095	Yes (18× faster)
MLP-only	∼2,500	0.99	Yes
Linearized TF	∼14,000	0.000	Yes (no Fourier)
Full transformer	∼37,000	0.10→0.54	Yes
Linear DMF	—	0.078	**No**
Softmax-only	—	0.103	**No**

Table 6: Grokking speed hierarchy. Nonlinearity is necessary (linear DMF never groks despite nuclear norm dropping from 7,340 to 3,877). Attention is not (DMF + ReLU groks 18× faster). Softmax alone is insufficient.

4.8 Continuous metrics reveal what binary IIA hides

Binary IIA treats a 51% probability shift the same as a 99% probability shift. For the always-Grassmannian operations where IIA saturates at 1.0, continuous metrics reveal important quality differences between methods. Full continuous metrics tables (KL, JS, probability difference, normalized logit difference) comparing DAS, NL-DAS, and Structured pi-SAE across all operations will be reported in the final version.

4.9 Surprising positive and negative cases

Cubic sum ($a^3 + b^3 \bmod 113$) achieves perfect equivariance (100%) despite cubing alone reaching only 50.9%. The boundary is not "group action versus polynomial" but whether the operation's binary structure provides sufficient additive symmetry.

Affine ($2a + 3b + 5 \bmod 113$) is a linear function that *fails* to produce a Grassmannian variable. Under an additive shift $a \mapsto a + g$, the output shifts by $2g$, not g. Equivariance requires the operation to be a group action, not merely a linear function.

Max ($\max(a, b) \bmod 113$) achieves 95.7% equivariance despite not being a group action. It is piecewise equivariant: when the argmax is stable under the shift, max behaves like a shifted identity.

5 The Mechanism: How Grokking Creates Causal Structure

The atlas (§4) established *what*: a sharp three-class partition governed by grokking. This section asks *how* and *why*. We trace the emergence of causal structure through five lenses: the minimal architecture required, the fine-grained dynamics of Fourier crystallization, spectral compression during the phase transition, the relationship between weight-space structure and activation-space causal variables, and the gauge-freedom of the causal subspace itself.

5.1 Minimal architecture for grokking

What is the simplest model that groks? We compare a depth-2 deep matrix factorization (DMF) $y = W_2 W_1 x$ against variants with nonlinearity (DMF + ReLU), a linearized transformer (identity attention + ReLU MLP), a softmax-only transformer (no MLP), and the full single-layer transformer. All are trained on modular multiplication ($\mathbb{Z}_{113}$) with weight decay 1.0 for 50,000 epochs.

Three findings emerge. First, **nonlinearity is necessary**: the linear DMF never generalizes despite 50,000 epochs. Its nuclear norm drops from 7,340 to 3,877—spectral compression occurs—but test accuracy remains ∼0%. Compression alone is not sufficient for grokking; the nonlinear activation function that enables the Fourier representation is essential.

Second, **attention is not necessary for grokking on modular multiplication**: DMF + ReLU groks 18× faster than the full transformer. The softmax attention mechanism adds enormous overhead to the grokking process without being required for the underlying algorithm on this operation. Whether this extends to other operations or multi-task settings remains open.

Epoch	Test accuracy	Fourier alignment
37,000	65.6%	0.070
37,500	91.3%	0.093
38,000	98.6%	0.143
38,500	99.7%	0.260
39,500	99.7%	0.498
40,000	99.8%	0.543

Table 7: Fine-grained Fourier trajectory for modular multiplication. Test accuracy crosses 90% at epoch 37,500; Fourier alignment crosses 0.30 approximately 1,000 epochs later. The model generalizes *before* its weights crystallize into the Fourier basis.

Epoch	Test acc.	AGOP FA	Weight FA	AGOP erank
5,000	0.4%	0.108	0.080	10.0
25,000	6.3%	0.712	0.081	9.8
30,000	19.1%	**0.894**	0.075	8.7
35,000	95.8%	0.662 (dip)	0.270	5.4
40,000	100%	**0.974**	0.864	4.7
45,000	100%	0.888	0.977	4.4

Table 8: AGOP trajectory for modular multiplication. The Fourier alignment of the gradient outer product shows a double-peak: 0.894 at epoch 30,000, followed by a dip to 0.662 during the grokking transition (epoch 35,000), then crystallization at 0.974 (epoch 40,000). The *gradient* discovers Fourier structure before the *weights* do (weight FA is 0.075 when AGOP FA is 0.894).

Third, the **linearized transformer groks without Fourier structure**: it achieves 100% test accuracy by epoch 14,000 but its Fourier alignment remains at 0.000 throughout. Grokking and Fourier representation are dissociable—grokking can occur via non-Fourier algorithms. The Fourier representation is one route to generalization, not the only one.

5.2 Fourier structure emerges after generalization

Prior work identified Fourier representations as the mechanism underlying grokking in modular arithmetic (Nanda et al., 2023; Zhong et al., 2023). A natural assumption is that Fourier structure *drives* grokking: the model discovers the Fourier basis, then generalizes. Our fine-grained trajectory analysis (1,002 checkpoints at 100-epoch resolution) reveals the opposite: **generalization precedes Fourier crystallization**.

Test accuracy crosses 90% at epoch 37,500, but Fourier alignment does not cross 0.30 until approximately epoch 38,500—a lag of $\sim$1,000 epochs. The model generalizes before its weights fully organize into the Fourier basis.

The AGOP double-peak. Tracking the Average Gradient Outer Product (AGOP) Fourier alignment through training reveals an unexpected non-monotonic pattern. For the full transformer:

The AGOP alignment peaks at 0.894, then *dips* to 0.662 during the actual grokking transition (epoch 35,000), before crystallizing at 0.974 post-grokking. The dip coincides with the period of rapid generalization—the phase transition temporarily disrupts the gradient's Fourier structure before it stabilizes. By contrast, the DMF's AGOP alignment remains flat at 0.06–0.08 throughout 50,000 epochs, confirming that the linear model never discovers Fourier structure even in its gradients.

The key implication: the gradient "knows" the Fourier basis $\sim$10,000 epochs before the weights do (AGOP FA is 0.894 at epoch 30,000 while weight FA is still 0.075). Grokking may be the process by which weight-space catches up to gradient-space structure.

Method	Site	Best k	IIA
SVD(FF)	attn_out	32	**0.999**
DAS	attn_out	8	0.150
DAS	mlp_out	8	0.631
SVD(FF)	mlp_out	64	0.444

Table 9: SVD of the feedforward product matrix vs. DAS on a grokked multiplication model. At the attn_out site, SVD achieves IIA = 0.999 *with no optimization* (DAS: 0.150 after 1,000 gradient steps). The Grassmannian distance between SVD and DAS subspaces is 1.5–8.2 radians (near-orthogonal).

Fourier structure lives in attention, not MLPs. In a grokked 1-layer model ($p = 113$), we measure Fourier selectivity across all components: 0 of 512 MLP neurons are Fourier-selective, and the embedding has a nearly flat Fourier spectrum (participation ratio = 3.93). The Fourier structure resides entirely in attention: the QK circuit selects individual frequency pairs (rank-1 structure), while the OV circuit operates in a broader k-dimensional subspace of Fourier modes. This decomposition is consistent with the "clock" interpretation of Zhong et al. (2023).

5.3 Spectral compression during grokking

The factored feedforward product matrix $W_{\text{FF}} = W_{\text{in}} W_{\text{out}}$ undergoes dramatic rank collapse during grokking. Tracking the effective rank (exponential of the Shannon entropy of the normalized singular value spectrum) through training:

Epoch	FF effective rank	Stage
34,000	48	Pre-grokking
37,500	14	Transition
41,000	9	Post-grokking

The 5.3× compression from effective rank 48 to 9 confirms that grokking compresses the model's representational capacity into a low-dimensional subspace. This compression is necessary but not sufficient: the linear DMF also achieves spectral compression (nuclear norm 7,340 → 3,877) without grokking (§5.1). What distinguishes successful grokking is that the compressed subspace acquires *structure*—specifically, equivariance under the operation's symmetry group.

5.4 Weight-space SVD predicts causal geometry

If grokking compresses the solution into a low-rank subspace, can we read the causal variable directly from the weight matrices without fitting DAS? We compare SVD of the feedforward product matrix against DAS fitted to activations, measuring IIA for both.

At the attention output site, SVD(FF) achieves IIA = 0.999 at $k = 32$ with *zero optimization*—no gradient steps, no interchange training—while DAS achieves only 0.150 at $k = 8$ after 1,000 gradient steps. The comparison is not dimension-matched: SVD requires 4× the subspace dimension, reflecting that the top singular vectors of W_{FF} include directions relevant to the causal variable but also spectral directions that are not causally active. Nevertheless, the fact that a zero-cost weight-space decomposition outperforms an optimized activation-space method is striking.

The Grassmannian distance between the SVD and DAS subspaces ranges from 1.5 to 8.2 radians—they are near-orthogonal. This is consistent with the gauge-freedom finding (§5.5): many different subspaces carry the same causal information. The SVD subspace is one valid parameterization; the DAS subspace is another. Neither is "correct"—both are equivalent points in the orbit of the gauge group acting on $\text{Gr}(k, d)$.

The site dependence is notable: SVD(FF) is most informative at attn_out (where the attention output has been computed) but weaker at mlp_out (where the MLP has further mixed the representation). DAS shows the opposite pattern, achieving 0.631 at mlp_out. This suggests that weight-space and activation-space

Operation	Still grokked?	Test loss	OV erank	Interpretation
Multiplication	No	20.4	9.19	Structure degraded
Comp. addition	**Yes**	9e-6	5.46	Structure preserved
Cubing	No	9.19	6.23	Higher erank

Table 10: Un-training results: retraining grokked models on wrong-operation labels. Composite addition retains its grokked structure (test loss 9e-6) even under wrong labels—the learned algorithm is robust to label corruption. Multiplication loses grokking (erank increases from the grokked baseline to 9.19).

methods have complementary strengths: SVD captures the structural potential of the weight matrices, while DAS captures the functional use of that structure under the data distribution.

5.5 Subspace identity is gauge freedom

If grokking produces a specific causal subspace, we might expect different random seeds to converge to the *same* point on $\mathrm{Gr}(k, d)$. They do not. Ten seeds of modular multiplication all grok successfully (IIA 0.94–1.00), but the recovered DAS subspaces are nearly orthogonal: pairwise overlap ≈ 0.008, where random k-dimensional subspaces of $\mathbb{R}^{128}$ would have overlap $k/d = 0.016$. The grokked subspaces are *less* aligned than chance.

The basin of attraction is enormous. Perturbing the DAS subspace by rotating it 1+ radians on $\mathrm{Gr}(k, d)$ barely drops IIA. The causal variable is not a fragile low-dimensional needle in a high-dimensional haystack—it is a robust feature of the representation that can be read out from exponentially many subspaces.

Spectral graph structure. Computing the pairwise Grassmannian distance between all 10 seeds' subspaces yields a nearly uniform distance matrix: $d_{\mathrm{Gr}} \in [1.93, 2.19]$. The subspaces are equidistant from each other, forming a "spectral simplex" on the Grassmannian rather than clustering around a preferred direction.

This finding has a concrete implication for mechanistic interpretability: **the identity of a causal subspace is not a meaningful property**. What matters is the equivariance of the learned function *within* the subspace, not the subspace itself. Two analyses of the same model at different random seeds would find orthogonal DAS solutions—but both solutions are equally valid because the model treats the choice of subspace as gauge freedom. This parallels the rotational gauge freedom in the factorized transformer's QK circuit, where $W_Q \to W_Q R$, $W_K \to W_K R$ leaves $W_Q W_K^\top$ invariant.

5.6 Structural robustness under un-training

If grokking creates causal structure, does un-training—continuing to train on the wrong operation's labels—destroy it? We present preliminary results (n=1 per condition) from three grokked models retrained with shuffled labels from a different operation.

Composite addition retains its grokked structure completely—test loss remains 9×10^{-6} and OV effective rank stays at 5.46. The learned algorithm is so deeply embedded in the weight structure that training on wrong labels cannot dislodge it. Multiplication, by contrast, loses grokking: its erank increases to 9.19, and test loss rises to 20.4.

The asymmetry is informative. Composite addition uses $p = 91$ (composite modulus), while multiplication uses $p = 113$ (prime). The composite modulus may create more redundant encoding of the Fourier structure, making it harder to un-learn. This connects to the stochastic grokking finding (§4.3): composite addition's grokking is seed-dependent, but once it grokks, the structure is more robust than prime-modulus operations.

5.7 Stratum classification: toy models vs. pretrained

We test whether the singular value structure of weight matrices—classifying each head's OV and QK circuits into strata based on spectral gap, cumulative variance, and effective rank—provides an alternative lens on causal geometry.

In grokked 1-layer models ($p = 113$), the classification works: all four heads have low-rank OV (erank 2.1–6.2) and rank-1 QK (gap ratio > 3), consistent with the Fourier mechanism where QK selects frequency pairs and OV operates in the active Fourier subspace.

In GPT-2-small, it fails completely. Of 144 heads, 143 fall into the moderate-rank stratum ($\mathcal{M}_\Sigma$) regardless of function. Circuit heads (IOI name movers, induction, s-inhibition) have *higher* average OV erank (54.6) than non-circuit heads (49.5)—the opposite of what low-rank structure would predict. The explanation is superposition: each head participates in multiple tasks, producing high effective rank even if each task-specific contribution is low-rank.

The methodological lesson: **functional diagnostics (equivariance) outperform structural diagnostics (spectral profile)** for classifying causal geometry. Equivariance cleanly separates the three-class partition without spectral analysis. Task-conditioned strata—projecting OV into a task-specific DAS subspace before classifying—may rescue the approach but remain untested.

6 Discussion

6.1 A hierarchy of causal validity

Our results establish a four-level hierarchy for evaluating causal variable claims, where each level adds a stronger requirement:

1. **Interchange effectiveness** (IIA; necessary, not sufficient): High IIA means the method supports interchange but does not distinguish genuine causal structure from memorization (squaring achieves IIA $= 1.0$ at $k = 4$ without grokking) or from degenerate encoder-decoders (NL-DAS achieves IIA $= 1.0$ at $k = 1$ via lookup tables).

2. **Interchange faithfulness** (diversity ratio, reconstruction; necessary for nonlinear methods): A faithful intervention modifies the causal variable while preserving all other information. The diversity ratio ρ (Eq. 5) and reconstruction MSE detect lookup-table failure modes. Without faithfulness, "high IIA" means only "the decoder can produce the right output," not "the encoder identified the right variable." Linear DAS is faithful by construction (the projection $QQ^\top$ preserves the orthogonal complement), but nonlinear methods must demonstrate faithfulness empirically.

3. **Distributional fidelity** (KL, JS, continuous metrics; distinguishes quality among effective methods): Among methods achieving IIA ≈ 1.0, continuous distributional metrics reveal how much of the clean distribution is preserved versus distorted. A method that flips the top-1 prediction (high IIA) while scrambling the rest of the distribution (high KL) is less valid than one that reproduces the full counterfactual distribution.

4. **Structural diagnostics** (equivariance, cross-distribution generalization): High equivariance means the recovered variable transforms coherently under the operation's symmetry group—the strongest evidence that the method has found a genuine causal variable rather than exploiting distributional artifacts. Cross-distribution generalization (disjoint names, novel templates) provides the analogous test for natural-language tasks that lack algebraic symmetries.

The hierarchy has an important asymmetry. Linear DAS automatically satisfies level 2 (faithfulness) because the orthogonal projection modifies only the subspace component, but it may fail level 1 when the causal variable is nonlinear. Unconstrained nonlinear methods easily satisfy level 1 but fail level 2—they can achieve perfect IIA by learning degenerate solutions. The Structured pi-SAE satisfies both because the ELBO provides the reconstruction and regularization constraints that prevent degeneracy.

6.2 Grokking enables causal structure—linear or nonlinear

The stochastic grokking discovery shows that grokking is the process by which a model's internal representation transitions from an unstructured memorization encoding to a structured causal encoding. For operations with linear algebraic structure, this structured encoding lives on $\mathrm{Gr}(k, d)$ and DAS recovers it. For operations with nonlinear structure, the encoding lives on a curved manifold and requires nonlinear methods to access.

The mechanism section (§5) reveals that this transition involves three dissociable processes: spectral compression (effective rank $48 \to 9$), Fourier crystallization (which *lags* generalization by $\sim$1,000 epochs), and the selection of one among exponentially many equivalent causal subspaces (gauge freedom). The AGOP analysis (§5.2) provides a concrete timeline: the gradient discovers Fourier structure $\sim$10,000 epochs before the weights do, and the actual phase transition is marked by a temporary *dip* in gradient-space Fourier alignment before crystallization.

The minimal architecture experiments (§5.1) further clarify: nonlinearity is necessary for grokking, but attention is not. DMF + ReLU groks 18$\times$ faster than the full transformer, and the linearized transformer groks without any Fourier structure at all. These findings suggest that Fourier representations are the dominant grokking mechanism for transformers with softmax attention, but grokking itself is a more general phenomenon—the transition from memorization to a structured causal encoding—that can occur through multiple algorithmic routes.

The key insight is that grokking is about *causal structure*, not *linear* causal structure. The Grassmannian perspective captures the linear case precisely, but the broader phenomenon is the emergence of any structured causal variable—linear or nonlinear—during the generalization transition.

6.3 Implications for language model interpretability

For practitioners applying DAS or nonlinear causal abstraction to language models:

1. **Never report IIA alone.** IIA is necessary but insufficient at every level: linear DAS on memorized models (squaring, IIA = 1.0) and unconstrained nonlinear methods (NL-DAS, IIA = 1.0) both achieve perfect IIA vacuously. Always report at least one faithfulness metric (diversity ratio for nonlinear methods, equivariance or cross-distribution generalization for any method).

2. **Use continuous distributional metrics.** KL divergence and normalized logit difference distinguish among methods that all achieve IIA $\approx$ 1.0. Binary IIA treats a 51% probability shift the same as 99%; continuous metrics capture this difference.

3. **Be suspicious of unconstrained nonlinear featurizers.** Any encoder-decoder trained end-to-end on interchange loss without reconstruction or regularization constraints can learn a lookup table. The ELBO (or an equivalent generative constraint) is necessary to prevent this failure mode.

4. **Test cross-distribution generalization.** For natural-language tasks, evaluate on disjoint entity sets or novel templates. A method that fails under distribution shift is exploiting surface regularities, not recovering causal structure.

5. **Consider nonlinear methods** when DAS IIA is low. Low IIA does not mean the causal variable does not exist—it may mean DAS is searching the wrong function class. But validate with faithfulness metrics.

6.4 Weight-space vs. activation-space methods

The SVD result (§5.4) reveals a surprising complementarity. Weight-space methods (SVD of $W_{\mathrm{in}}W_{\mathrm{out}}$) achieve near-perfect IIA at attn_out *without any optimization*, outperforming 1,000 steps of DAS gradient descent. This suggests that for grokking models, the causal variable is directly legible in the weight matrices—DAS's optimization is recovering structure that was already "written down" in the weights.

The practical implication is that weight-space analysis should precede activation-space methods: if SVD of the relevant weight product achieves high IIA, DAS optimization may be unnecessary. If SVD fails but DAS succeeds, the discrepancy suggests the causal variable involves computation beyond simple linear readout from the weight matrices.

6.5 Limitations

Our grokking experiments use single-layer transformers on synthetic tasks. Whether the grokking–Grassmannian link holds in deeper models or natural language is an open question, though the k-sweep and equivariance diagnostics apply regardless. The Structured pi-SAE introduces a model-selection problem (architecture, β, α) that DAS avoids. Multi-seed stability experiments (10 seeds per stochastic operation) are underway. The connection between the Structured pi-SAE latent space and the Fourier representation structure identified by Nanda et al. (2023) deserves further theoretical analysis. The stratum classification (§5.7) fails on pretrained models; task-conditioned strata remain untested. The un-training experiments (§5.6) are preliminary (n=1 per condition, three operations)—multi-seed replication and broader operation coverage are needed before drawing firm conclusions about un-training robustness.

7 Related Work

Causal abstraction. Causal abstraction provides the theoretical foundation for DAS (Geiger et al., 2021; 2023). Wu et al. (2023) scaled DAS via Boundless DAS. Makelov et al. (2024) identified interpretability illusions where dormant pathways inflate IIA. Our geometric diagnostics address a complementary problem: even within the linear intervention class, high IIA does not imply genuine linear structure.

Grokking. Grokking was introduced by Power et al. (2022) and mechanistically analyzed through Fourier representations (Nanda et al., 2023; Zhong et al., 2023). Liu et al. (2023) frame grokking as compression; our spectral analysis (§5.3) gives this compression concrete numbers (effective rank $48 \to 9$, 5.3×) and shows it is necessary but not sufficient—linear DMFs compress without grokking. Our AGOP analysis (§5.2) adds temporal resolution absent from prior work: the gradient discovers Fourier structure ~10,000 epochs before the weights, and generalization precedes Fourier crystallization by ~1,000 epochs. Our contribution is to show that grokking has a specific consequence for causal abstraction: it is the process that converts non-Grassmannian representations into Grassmannian ones (for linear operations) or into structured nonlinear ones (for nonlinear operations). Stochastic grokking has been observed via the slingshot mechanism (Thilak et al., 2022) but not connected to causal geometry.

Spectral analysis of neural networks. Random matrix theory has been applied to weight matrices for understanding training dynamics (**?**). Our stratum classification (§5.7) extends this to a functional taxonomy, but reveals an important limitation: spectral profiling has discriminative power for single-task models (where it correctly identifies QK as rank-1 and OV as low-rank in grokked models) but loses all discriminative power under superposition (143/144 GPT-2 heads are $\mathcal{M}_\Sigma$ regardless of function). Task-conditioned spectral analysis may bridge this gap but remains untested.

Disentangled representations and causal variables. Narayanaswamy et al. (2017) introduced semi-supervised VAEs for disentanglement. Recent work on identifiable representations (Khemakhem et al., 2020) establishes conditions under which latent factors can be recovered. Locatello et al. (2019) proved that unsupervised disentanglement is impossible without inductive biases; our structured prior is precisely the auxiliary supervision their theorem requires. Our application differs in that we evaluate disentanglement through causal interchange (IIA), not reconstruction quality, connecting the VAE literature to causal abstraction.

Representation geometry. Grassmannian methods have been applied to subspace tracking and domain adaptation (Edelman et al., 1998; Gopalan et al., 2011). Linear representation hypotheses in interpretability (Marks & Tegmark, 2023; Engels et al., 2025; Park et al., 2024) argue that many concepts are encoded in linear subspaces; our work tests this hypothesis causally rather than correlationally.

Nonlinear causal abstraction. Concurrent work on nonlinear interchange interventions (Geiger et al., 2024) uses neural network featurizers before applying DAS. Our results show that this approach is vulnerable to the lookup-table failure mode (§4.7): unconstrained featurizers achieve perfect IIA by learning degenerate encoder-decoders. Our Structured pi-SAE approach uses the ELBO to provide the reconstruction and regularization constraints that prevent this degeneracy, connecting to the identifiability literature (Khemakhem et al., 2020; Sutter et al., 2020) which establishes that auxiliary supervision is necessary for recovering latent factors. The NL-DAS failure mode and our faithfulness metrics provide empirical evidence for this theoretical requirement in the causal abstraction setting.

8 Conclusion

DAS searches for causal variables in a specific function class: linear subspaces on the Grassmannian. Our atlas of 14 operations maps where this search succeeds and where it fails. The boundary is sharp, governed by grokking, and stochastic for operations near the transition.

Three practical implications follow. First, IIA should never be reported alone: IIA = 1.000 is compatible with a genuine causal variable, a memorized lookup table, and a degenerate encoder-decoder. The four-level hierarchy we propose—interchange effectiveness, interchange faithfulness, distributional fidelity, and structural diagnostics—provides a complete evaluation framework.

Second, nonlinear extensions of causal abstraction require generative structure and structured sparsity. Unconstrained nonlinear featurizers achieve perfect IIA vacuously; the Structured pi-SAE's combination of ELBO, label-conditional prior, and L_1 sparsity prevents this failure mode. End-to-end intervention training closes the gap between training and evaluation objectives, achieving IIA = 0.97 on GPT-2 hypernymy where DAS gets 0.07. This connects causal abstraction to the identifiability literature: auxiliary supervision plus generative constraints are necessary for valid nonlinear causal variable recovery.

Third, continuous distributional metrics (KL, JS, normalized logit difference) should replace binary IIA as the primary evaluation measure. Binary IIA saturates too easily and hides important quality differences between methods.

Fourth, the mechanism underlying grokking's creation of causal structure is more nuanced than previously understood. Nonlinearity is necessary but attention is not (DMF + ReLU groks 18× faster). Fourier structure *follows* generalization rather than driving it, and the gradient discovers this structure ∼10,000 epochs before the weights crystallize. The causal subspace itself is gauge freedom—10 seeds produce orthogonal subspaces that all support equally valid causal interventions. And spectral profiling, while informative for single-task models, fails completely under superposition. These findings suggest that *functional* diagnostics (equivariance, intervention faithfulness) are more robust than *structural* diagnostics (spectral profile, stratum classification) for evaluating causal geometry.

The linearity assumption underlying DAS is not a benign default. It is a testable hypothesis. And the natural fix—adding nonlinearity—introduces a new failure mode that is invisible to the standard evaluation metric. This paper provides the diagnostics to detect both problems, the mechanistic analysis to understand why they arise, and the structured nonlinear method that avoids them.

Data and code. All code for model training, DAS fitting, structured VAE training, geometric diagnostics, and figure generation is available at `[repositoryURL]`.

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